Smart Programming : YouTube Channel

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JDBC (PART 16)

Connection Pooling

How many ways are there to get connection object

1. DriverManager (Class) - java.sql package
2. DataSource (Interface) - javax.sql package

Using DriverManager, database connection follows below life cycle:-

1. Opening a connection to the database using the database driver (connection object created)
2. Opening a TCP socket for reading or writing data

3. Reading or writing data over the socket
4. Close the connection
5. Close the socket

- Getting connection object using DriverManager are expensive operations
- Creating a physical connection is time consuming, thus the server mentains a pool of available connections to increase the performance

Drawbacks of DriverManager

1. DriverManager takes a lot of time (even 1s) to open database conneciton in a network which can slow down the performance of java application

2. Whenever there is increase in no of clients, the performance of java application will get decreased

3. The connection object created using DriverManager is not reusable and due to this there is performance issue with server

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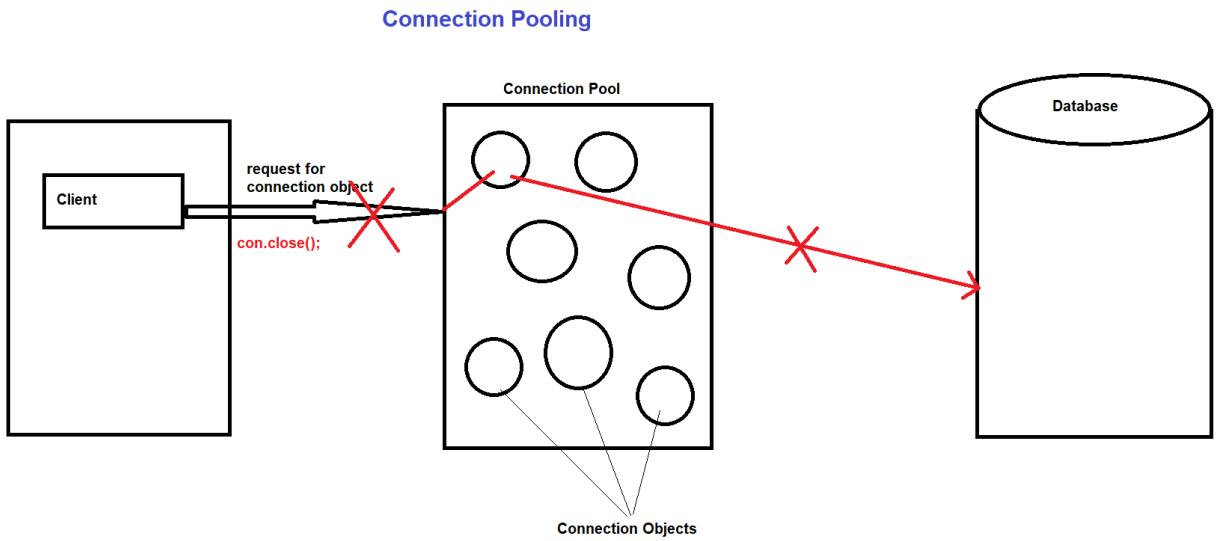
What is connection pooling ?

- A connection pool is a group of reusable connection objects for a perticular database
- When an application requests a connection, it obtains one from pool.
- When an application closes the connection, the connection object returns back to the pool



How connection pooling works (SERVER)

1. A server administrator prepares a connection pool with a set off connections and also a mediator object as DataSource object to access the pool

2. An administrator stores DataSource object into JNDI registry

3. A java application reads DataSource object from JNDI and requests for a connection object from connection pool

4. The DataSource object takes the connection object, performs its tasks and send the connection object back to the pool

What is DataSource

```
interface DataSource
{
   getConnection()
   getConnection(String username, Stirng
password)
}
```

- DataSource is a factory for connections to the physical data source that this DataSource object represents.
- An alternative to the DriverManager facility, a DataSource object is the preferred means of getting a connection.
- An object that implements the DataSource interface will typically be registered with a naming service based on the JavaTM Naming and Directory (JNDI) API.

Different ways to provide Connection pooling

1. Default connection pooling provided by database vendors (no need to use external jar files)

2. Third party connection pooling mechanism

(we have to use external jar files)

a. Apache Commons DBCP (Database Connection Pooling)

b. Tomcat JDBC

c. C3P0

d. Proxool

e. BoneCP

f. Hikary

3. Application server provided connection pooling

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1. Default connection pooling provided by database vendors

(no need to use external jar files)

Steps for Connection pooling

1. Create DataSource object

   for MySQL - MySQLDataSource

   for Oracle - OracleDataSource

2. Set connection properties to the DataSource

   Methods used are

   -> setURL(Stirng driver_url)

   -> setUser(String username)

   -> setPassword(String password)

3. Get connection object form DataSource

-> getConnection()

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